NAME:…… …… FORM 4……..

ST MARY'S SECONDARY SCHOOL

2021 MALAWI SCHOOL CERTIFICATE OF EDUCATION MOCK EXAMINATION

MATHEMATICS

Subject number:M131/II

Friday, 06th August 2021

Time allowed: 2 h 30 mins
08:00 – 10:30 am

PAPER II
(100 marks)

Instructions:

1. **This paper contains 12 printed pages. Please check.**
2. Answer **all** the **six** questions in Section **A** and any **four** questions in Section **B**.
3. The maximum number of marks for each answer is indicated against each question.
4. Write your answers in the spaces provided on the question paper.
5. Scientific calculators may be used.
6. **All working must be clearly shown.**
7. Write your **Full name** at the top of question paper in the spaces provided.
8. In the table provided on this page, **tick** against the question number you have answered.

Question number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
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7			
8			
9			
10			
11			
12			

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Section A (60 marks)

Answer all the **six** questions in this section

1. a. A point $\boldsymbol{A(-1,3)}$ is translated to $\boldsymbol{A'(3,8)}$. Calculate the translation in column vector **(3 marks)**

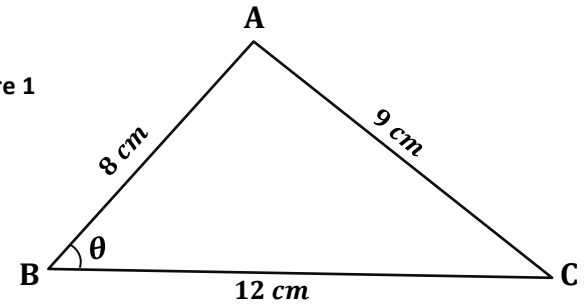
b. Simplify $\frac{3}{a-b} - \frac{a-3b}{(a-b)^2}$ **(5 marks)**

2. a. Given that $\log 3 = m$, express $\log\left(\frac{3}{\sqrt{3}}\right)$ in terms of m (**4 marks**)

b. The scale factor of two similar shapes is $\mathbf{\frac{2}{3}}$. The area of the larger shape is $117\ cm^2$. Find the area of the smaller shape. (**4 marks**)

3. a. **Figure 1** shows a triangle ABC in which $\boldsymbol{AB} = 8\ cm,\ \boldsymbol{BC} = 12\ cm, \boldsymbol{AC} = 9\ cm$ and angle $\boldsymbol{ABC} = \boldsymbol{\theta}$

Figure 1



Calculate the value of $\boldsymbol{\theta}$, leaving your answer correct to three significant figures

(6 marks)

b. When $\boldsymbol{x^3 + hx^2 + x - 4}$ is divided by $\boldsymbol{x - 1}$ the remainder is 2. Using the remainder theorem, calculate the value $\boldsymbol{h}$. **(4 marks)**

4. a. Prove the theorem which states that the angle between a chord and a tangent is equal to the angle subtended by the chord in the alternate segment. **(6 marks)**

b. Table 1 shows the ages and their frequencies of girls in a choir

Age in years	14	15	16	17
Frequency	3	4	5	4

Table 1

Find the mean age of the girls in the choir, giving your answer correct to 1 decimal place. **(5 marks)**

5. a. Solve the following equations simultaneously:

$$\boldsymbol{x + y = 3}$$
$$\boldsymbol{x^2 - y^2 = -3}$$

(6 marks)

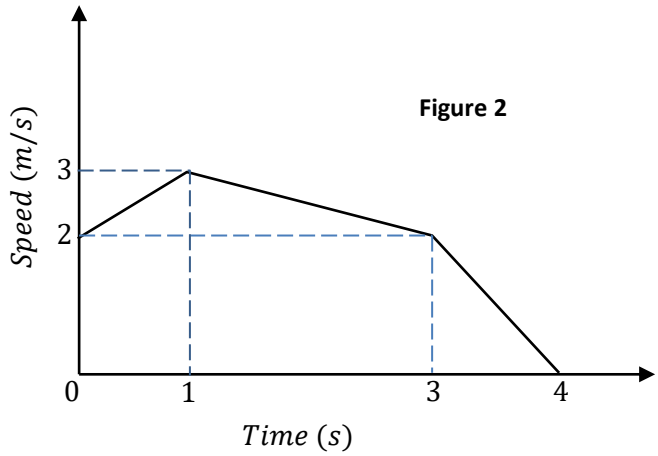
b. (i). Using a ruler and a pair of compasses only, construct in the same diagram:

- a circle centre **O** of radius 3 cm
- two tangents to the circle at **P** and **Q** from an external point **T** which is 8 cm away from the centre **O**

(ii). Measure and state the value of angle PTQ **(6 marks)**

6. a. The first term of an Arithmetic Progression (AP) is 3 and its last term is 47. If the sum of the AP is 575, calculate the common difference **(6 marks)**

b. Figure 2 is a sketch of a speed – time graph of a moving object



Calculate the distance covered by the object in the last 3 seconds **(5 marks)**

Section B (40 marks)

Answer any **four** questions from this section.

7. Table 2 shows masses in kg of 40 students

59	54	51	56	59	61	60	61
59	58	62	61	63	64	58	57
56	60	62	60	61	65	58	57
54	52	62	67	69	49	56	58
60	60	62	58	51	57	70	63

Table 2

a. Using data in table 1 and taking class intervals of $46 - 50$, $51 - 55$, … construct a frequency distribution table for the data

b. Using a scale of 2 cm to represent 5 units on the vertical axis and 2 cm to represent a class on the horizontal axis, draw a histogram for the data. **(10 marks)**

8. The Cost (C) of making a table varies partly as the area (A) of the table and partly as the square of length (L) of the table. If a table, $3m$ by $2m$ cost $K60\ 000$ and another table, $2m$ by $1m$ costs $K25\ 000$, find the cost of a table that is $4m$ by $5m$. **(10 marks)**

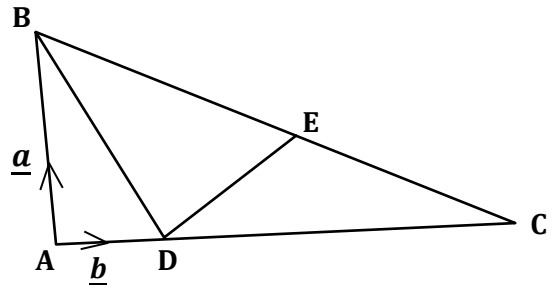
9. Solve the equation $n^3 + n^2 - 9n - 9 = 0$ **(10 marks)**

10. Mrs Banda want to spend K15 000 on bottles of cooking oil and tablets of soap. A bottle of cooking oil cost K1 500 each and a tablet of soap cost K600 each. She decides to buy at least 2 bottles of cooking oil, at least 5 tablets of soap and spend more on cooking oil than soap by at least K3 000.
    i. If $\boldsymbol{x}$ represents the number of bottles of cooking oil and $\boldsymbol{y}$ represents number of tablets of soap, write down **three** inequalities that satisfy the above information in addition to $\boldsymbol{x} \geq \mathbf{2}$.
    ii. Use a scale of 2 cm to represent 2 units on the horizontal axis and 2 cm to represent 5 units on the vertical axis, draw graphs to show the region bounded by the inequalities, shade the unwanted region.
    iii. Using the graph, find the maximum number of bottles of cooking oil she can buy.

(10 marks)

11. Figure 3 show triangle ABC. D divide side AC in the ratio 1: 4 and E is the midpoint of BC. $\overrightarrow{AB} = \underline{\boldsymbol{a}}$ and $\overrightarrow{\mathrm{AD}} = \underline{\boldsymbol{b}}$

Figure 3



Express $\overrightarrow{DE}$ in terms of $\underline{\boldsymbol{a}}$ and $\underline{\boldsymbol{b}}$ **(10 marks)**

12. The **table 3 below** shows some values of x and y for the equation $\boldsymbol{y = x^3 - 6x - 1}$

x	-3	-2	-1	0	1	2	3
y	-10		4	-1	-6	-5	

Table 3

i. Complete the table of values

ii. Using a scale of $2\ cm$ to represent 1 unit on the horizontal axis and $2\ cm$ to represent 5 units on the vertical axis, draw the graph of $\boldsymbol{y = x^3 - 6x - 1}$ on the graph paper provided.

iii. Use your graph to solve the equation $\boldsymbol{x^3 - 8x + 3 = 0}$ **(10 marks)**

END OF QUESTION PAPER.